

Key Sections for Final Review

Chapter 1:

1.3 Data Abstraction and Encapsulation

1.5 Algorithm Specification

1.7 Performance Analysis and Measurement

补充：数据的结构分为**逻辑结构**和**物理存储结构**。

数据的存储结构分为：**顺序存储结构**、**链式存储结构**、**索引存储结构**、**散列存储结构**

Chapter 2:

2.2 The Array as an Abstract Data Type

2.3 The Polynomial Abstract Data Type

2.4 Sparse Matrices

2.6 The String Abstract Data Type—KMP算法

Chapter 3 :

3.2 The Stack Abstract Data Type

3.3 The Queue Abstract Data Type

3.5 A Mazing Problem

3.6 Evaluation of Expressions

Chapter 4:

4.1 Singly Linked Lists and Chains

4.2 Representation Chains in C++

4.4 Circular Lists

4.5 Available Space Lists

4.6 Linked Stacks and Queues

4.7 Polynomials

4.9 Sparse Matrices

4.10 Doubly Linked Lists

~~4.11 Generalized Lists~~

Chapter 5:

5.1.1 Terminology of Trees

5.1.2 Representation of Trees

5.2 Binary Trees

5.3 Binary Tree Traversal and Tree Iterators

5.4 Additional Binary Tree Operations

5.5 Threaded Binary Trees

5.6 Heaps

5.7.1—4 Binary Search Trees

5.8 Selection Trees—胜者树、败者树的构建是重中之重

5.9 Forests

5.10 Representation of Disjoint Sets

Chapter 6:

6.1.2 Definitions (Graph)

6.1.3 Graph Representations

6.2.1 Depth-First Search

6.2.2 Breadth-First Search

6.2.4 Spanning Trees

6.3.1 Kruskal's Algorithm

6.3.2 Prim's Algorithm

6.4.1 Single Source/All Destinations:

Nonnegative Edge Costs

6.4.3 All-pairs Shortest Paths

6.5.1 Activity-on-Vertex (AOV) Networks

6.5.2 Activity-on-Edge (AOE) Networks

Chapter 7:

7.2 Insertion Sort (希尔排序思想要清楚)

7.3 Quick Sort

7.5 Merge Sort

7.6 Heap Sort

7.7 Sorting on Several Keys

7.9 Summary of Internal Sorting

7.10.2 k-Way Merging

7.10.5 Optimal Merging of Runs(Huffman's Algorithm)

Chapter 8:

8.2.1 Hash Tables

8.2.2 Hash Functions

8.2.4 Overflow Handling

Chapter 10:

10.2 AVL Trees—AVL树的构建和调整是重中之重

Chapter 11:

11.1 m-Way Search Trees

11.2 B-Trees

11.3 B⁺-Trees